Ex:No: 6A Date: 08/04/26	Customer Purchase Category Prediction Using Naive Bayes Classifier

Problem Description:

An e-commerce company wants to predict whether a customer will make a **High** or **Low** purchase based on demographic and behavioral attributes. The dataset contains both **categorical** (Gender, Membership) and **numerical** (Age, Visits per Month) features.

To perform probabilistic classification and handle mixed data types efficiently, the **Naive Bayes classifier** is used.

Objective:

To implement the Naive Bayes classification algorithm in R by training a model on customer data, predicting purchase categories, and evaluating classification accuracy.

Procedure:

- **Step 1: Create Dataset** - customer_data

	Age	Visits	Gender	Membership	Purchase
1	22	2	Male	Regular	Low
2	35	5	Female	Gold	High
3	40	6	Female	Gold	High
4	28	3	Male	Regular	Low
5	50	8	Female	Gold	High
6	45	7	Male	Silver	High
7	30	4	Female	Silver	Medium
8	55	9	Male	Gold	High
9	27	3	Male	Regular	Low
10	38	6	Female	Silver	Medium

```
customer_data <- data.frame(
  Age=c(22, 35, 40, 28, 50, 45, 30, 55, 27, 38),
  visits= c (2, 5, 6, 3, 8, 7, 4, 9, 3, 6),
  Gender = c("Male", "Female", "Female", "Male", "Female", "Male",
             "Female", "Male", "Male", "Female"),
  Membership = c ("Regular", "Gold", "Gold", "Regular", "Gold",
                  "Silver", "Silver", "Gold", "Regular", "Silver"),
  Purchase = c ("Low", "High", "High", "Low", "High", "High",
                "Medium", "High", "Low", "Medium")
)
customer_data
```

> customer_data

Age	Visits	Gender	Membership	Purchase
22	2	Male	Regular	Low
35	5	Female	Gold	High
40	6	Female	Gold	High
28	3	Male	Regular	Low
50	8	Female	Gold	High
45	7	Male	Silver	High
30	4	Female	Silver	Median
55	9	Male	Gold	High
27	3	Male	Regular	Low
38	6	Female	Silver	Medium

- **Step 2: Convert Categorical Variables to Factors**

```
customer_data$Gender <- as.factor(customer_data$Gender)
customer_data$Membership <- as.factor(customer_data$Membership)
customer_data$Purchase <- as.factor(customer_data$Purchase)
```

- **Step 3: Train(70%)–Test Split(30%)**

```
index <- sample(1:nrow(customer_data), 0.70*nrow(customer_data))
train_data <- customer_data[index,]
test_data <- customer_data[-index,]
```

- **Step 4: Apply Naive Bayes Algorithm**

#install packages "e1071" and load

```
install.packages("e1071")
library(e1071)
nb_model <- naiveBayes(Purchase~., data = train_data)
nb_model.
```

> nb_model

Naive Bayes classifier for Discrete Predictions

Call:

naiveBayes.default(x = X, y = Y, laplace = laplace)

A-priori probabilities:

Y

High	Low	Medium
0.5714286	0.2857143	0.1428571

Conditional probabilities:

Age

Y	[,1]	[,2]
High	46.25	8.539126
Low	25.00	4.242641
Medium	30.00	NA

visits

Y	[,1]	[,2]
High	7.25	1.7078251
Low	2.50	0.7071068
Medium	4.00	NA

Gender

Y	Female	Male
High	0.5	0.5
Low	0.0	1.0
Medium	1.0	0.0

Membership

Y	Gold	Regular	Silver
High	0.75	0.00	0.25
Low	0.00	1.00	0.00
Medium	0.00	0.00	1.00

- **Step 5: Predict on Test Data**

```
Predicted_purchase <- predict(nb_model, test_data)
predicted_purchase
```

- **Step 6: Confusion Matrix and Accuracy**

```
confusion_matrix <- table(
        Actual = test_data$Purchase
        Predicted = predicted_purchase
)
confusion_matrix
accuracy <- sum(diag(confusion_matrix))/sum
                                (confusion_matrix)
accuracy.
```

- **Step 7: Data Visualization for Naive Bayes Classification**
  - **Class Distribution Plot**
    - Shows how many customers belong to each purchase category.
    - Helps identify class imbalance.

```
barplot(table(customer_data$Purchase),
col= c("Skyblue", "orange", "lightgreen"),
main = "Purchase category distribution",
xlab= "Purchase category",
ylab= "Number of customers")
```

> Predicted_purchase

[1] High Low High

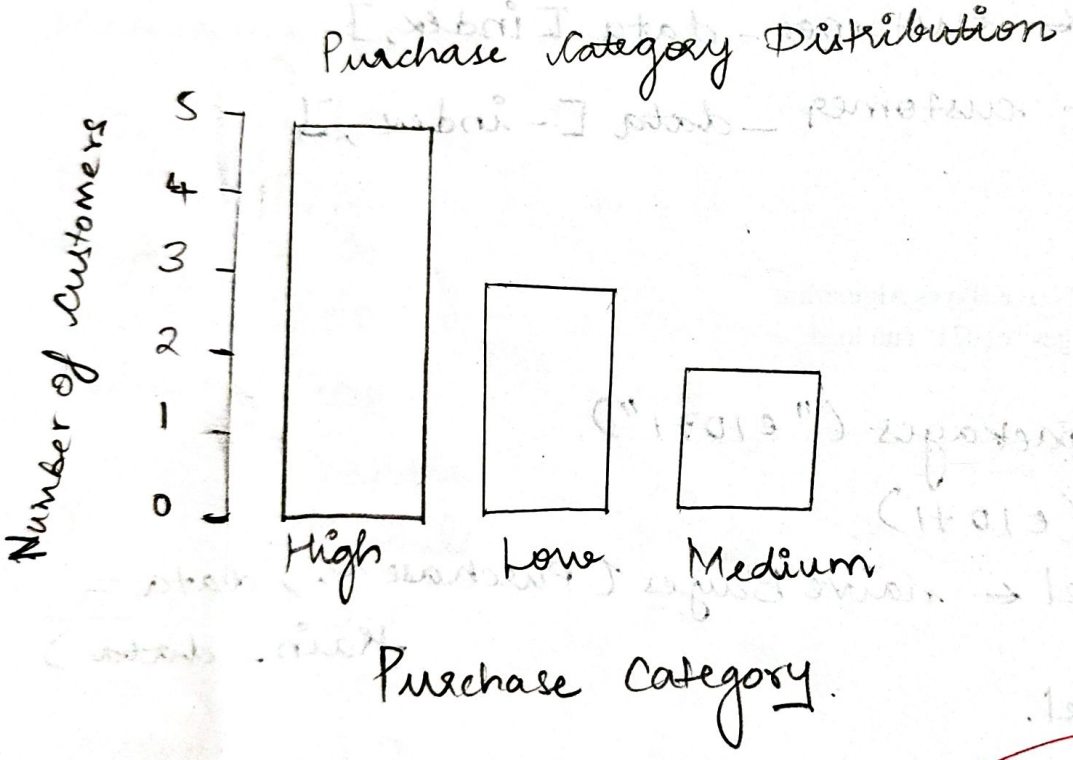
Levels: High Low Medium

> Confusion_matrix

Actual	Predicted High	Low	Medium
High	1	0	0
Low	0	1	0
Medium	1	0	0

> accuracy

[1] 0.6666667



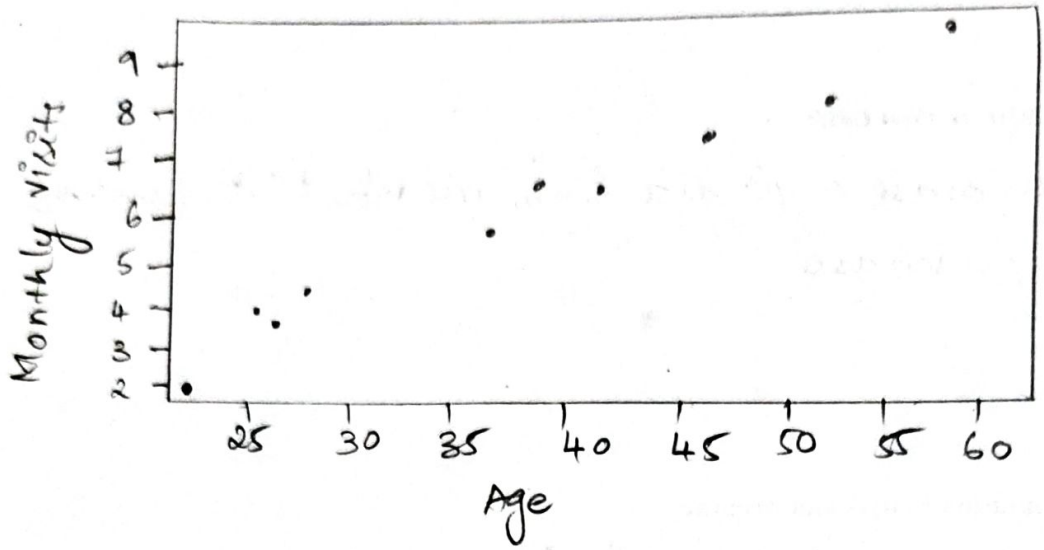
- **Scatter Plot (Age vs Visits)**
  - Displays customer grouping based on two numerical features – Age and Visits
  - Colors represent purchase categories.

```
plot(customer_data$Age, customer_data$Visits,
  col = as.numeric(customer_data$Purchase),
  pch = 19,
  xlab="Age",
  ylab="monthly visits",
  main = "Customer Distribution by Purchase Category")
  legend("topleft",
      legend = levels(customer_data$Purchase),
      Col = 1:length(levels(customer_data$Purchase)),
      pch = 19)
```

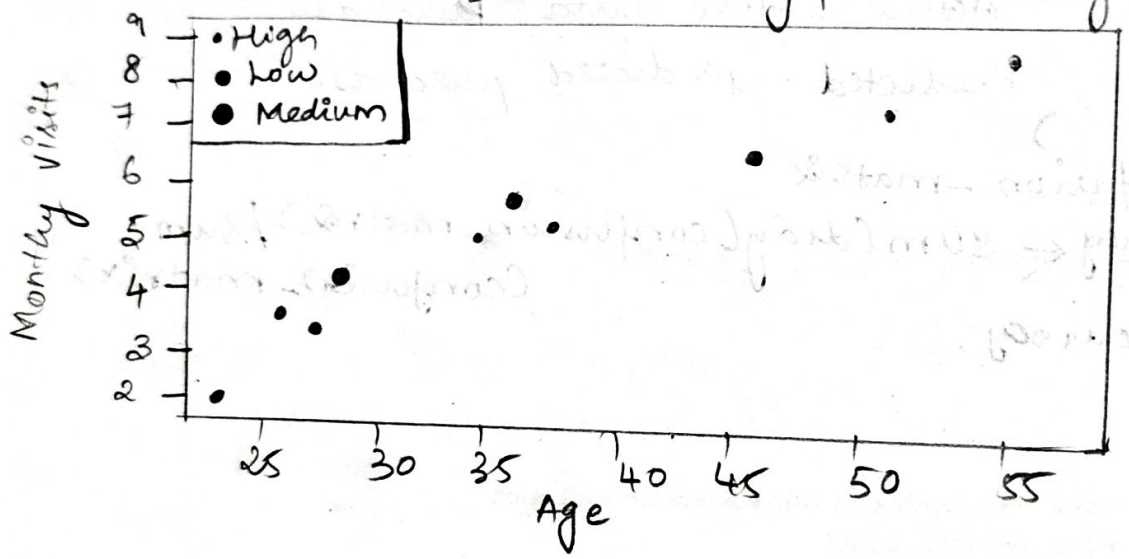
- **Boxplot for Numerical Attributes - Age and Purchase Category**

```
boxplot(Age ~ Purchase, data = customer_data,
  col = c("pink", "lightblue", "lightgreen"),
  main = "Age Distribution by Purchase Category",
  xlab= "Purchase category",
  ylab= "Age")
```

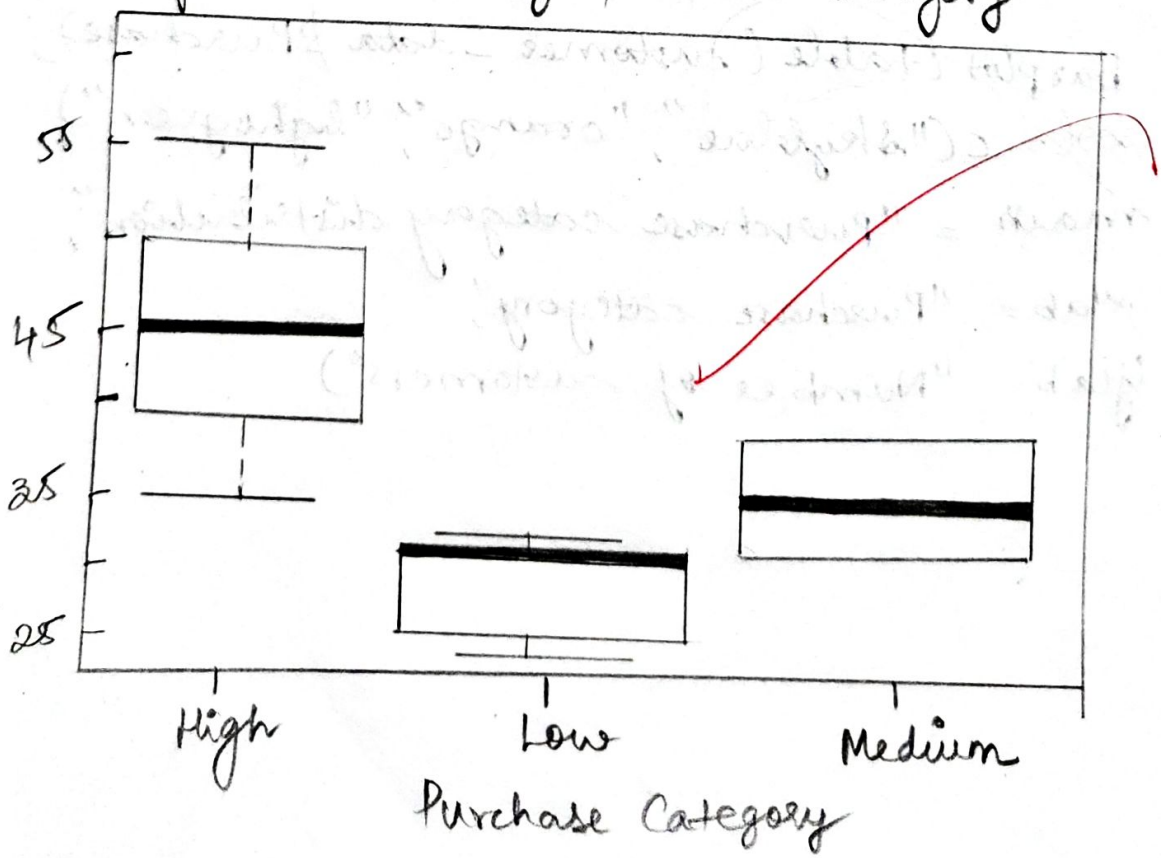
Customer Distribution by Purchase Category



Customer Distribution by purchase Category



Age Distribution by purchase category



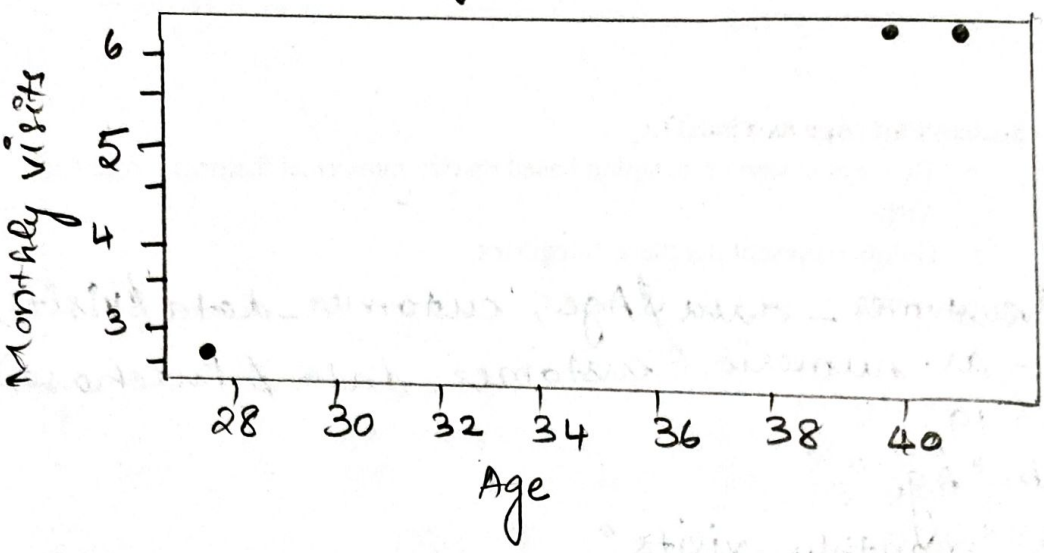
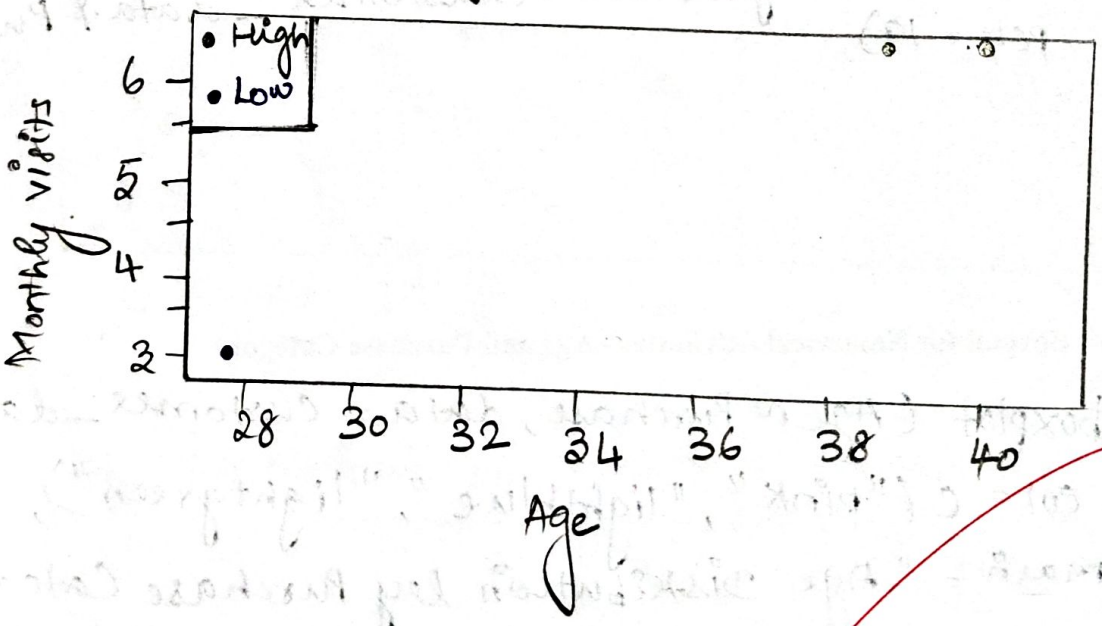
- **Visualization of Prediction Results**

```
plot (test_data$Age, test_data$visits,
col= as.numeric (predicted_purchase),
pch =19,
xlab = "Age",
ylab = " Monthly visits",
main = "Naive Bayes Classification Results")
legend ("topleft",
        legend = levels( predicted_purchase),
        col= 1:length (levels (predicted_purchase)),
        pch = 19)
```

Result:

Thus, the Naive Bayes classifier was successfully implemented in R to predict customer purchase categories using both numerical and categorical attributes. The model effectively applies probabilistic reasoning and produces accurate classification results, making it suitable for customer behavior prediction in e-commerce applications.

Naive Bayes Classification Results



Naive Bayes Classification Results